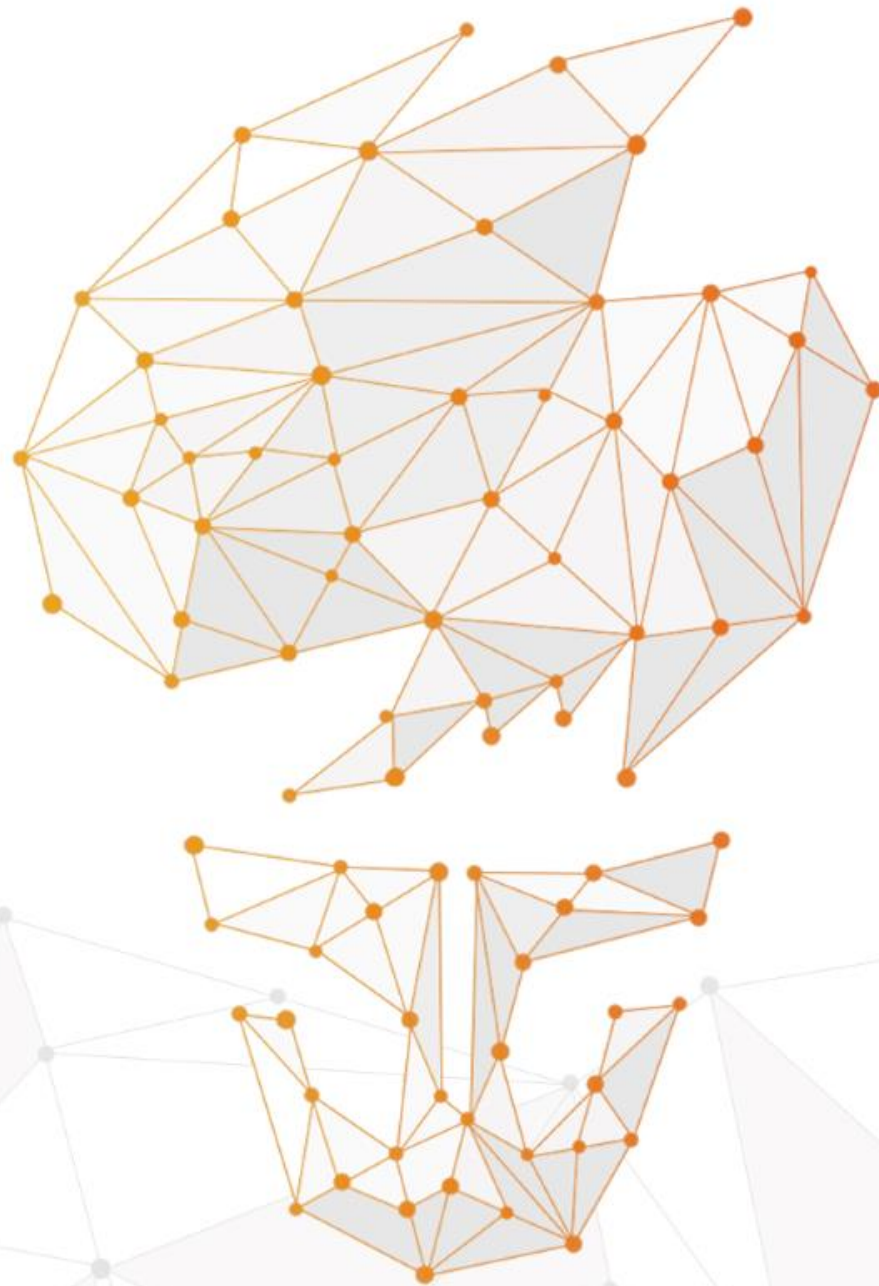


高数期中复习

致诚书院学长学姐带飞课

Chapter 10, Chapter 11

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Chapter10

无限数列与无穷级数

数列极限的定义

DEFINITIONS The sequence $\{a_n\}$ **converges** to the number L if for every positive number ϵ there corresponds an integer N such that for all n ,

$$n > N \quad \Rightarrow \quad |a_n - L| < \epsilon.$$

If no such number L exists, we say that $\{a_n\}$ **diverges**.

If $\{a_n\}$ converges to L , we write $\lim_{n \rightarrow \infty} a_n = L$, or simply $a_n \rightarrow L$, and call L the **limit** of the sequence (Figure 10.2).

DEFINITION The sequence $\{a_n\}$ **diverges to infinity** if for every number M there is an integer N such that for all n larger than N , $a_n > M$. If this condition holds we write

$$\lim_{n \rightarrow \infty} a_n = \infty \quad \text{or} \quad a_n \rightarrow \infty.$$

Similarly if for every number m there is an integer N such that for all $n > N$ we have $a_n < m$, then we say $\{a_n\}$ **diverges to negative infinity** and write

$$\lim_{n \rightarrow \infty} a_n = -\infty \quad \text{or} \quad a_n \rightarrow -\infty.$$

极限与初始项无关！

1. 数列极限的定义未给出求极限的方法；
2. 极限的精确定义不做要求。

$$(a) \quad \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

数列极限的运算法则

THEOREM 1 Let $\{a_n\}$ and $\{b_n\}$ be sequences of real numbers, and let A and B be real numbers. The following rules hold if $\lim_{n \rightarrow \infty} a_n = A$ and $\lim_{n \rightarrow \infty} b_n = B$.

- | | |
|-----------------------------------|---|
| 1. <i>Sum Rule:</i> | $\lim_{n \rightarrow \infty} (a_n + b_n) = A + B$ |
| 2. <i>Difference Rule:</i> | $\lim_{n \rightarrow \infty} (a_n - b_n) = A - B$ |
| 3. <i>Constant Multiple Rule:</i> | $\lim_{n \rightarrow \infty} (k \cdot b_n) = k \cdot B$ (any number k) |
| 4. <i>Product Rule:</i> | $\lim_{n \rightarrow \infty} (a_n \cdot b_n) = A \cdot B$ |
| 5. <i>Quotient Rule:</i> | $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{A}{B}$ if $B \neq 0$ |

数列极限 VS 函数极限

函数极限的运算法则

THEOREM 1—Limit Laws If L, M, c , and k are real numbers and

$$\lim_{x \rightarrow c} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow c} g(x) = M, \quad \text{then}$$

1. *Sum Rule:* $\lim_{x \rightarrow c} (f(x) + g(x)) = L + M$

2. *Difference Rule:* $\lim_{x \rightarrow c} (f(x) - g(x)) = L - M$

3. *Constant Multiple Rule:* $\lim_{x \rightarrow c} (k \cdot f(x)) = k \cdot L$

4. *Product Rule:* $\lim_{x \rightarrow c} (f(x) \cdot g(x)) = L \cdot M$

5. *Quotient Rule:* $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{L}{M}, \quad M \neq 0$

6. *Power Rule:* $\lim_{x \rightarrow c} [f(x)]^n = L^n, \quad n \text{ a positive integer}$

7. *Root Rule:* $\lim_{x \rightarrow c} \sqrt[n]{f(x)} = \sqrt[n]{L} = L^{1/n}, \quad n \text{ a positive integer}$

(If n is even, we assume that $\lim_{x \rightarrow c} f(x) = L > 0$.)

1. 有 + 无 = 无

2. 无 + 无 = 不定

3. 有 * 无 = 不定

4. 无 * 无 = 不定

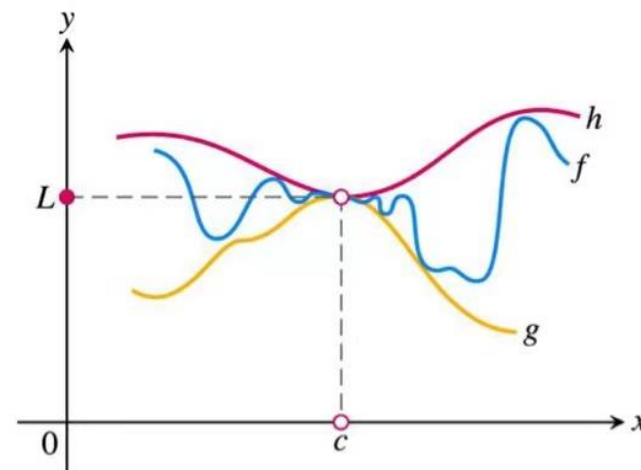
Sandwich Theorem

THEOREM 2—The Sandwich Theorem for Sequences Let $\{a_n\}$, $\{b_n\}$, and $\{c_n\}$ be sequences of real numbers. If $a_n \leq b_n \leq c_n$ holds for all n beyond some index N , and if $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$, then $\lim_{n \rightarrow \infty} b_n = L$ also.

(a) $\frac{\cos n}{n} \rightarrow 0$

(b) $\frac{1}{2^n} \rightarrow 0$

(c) $(-1)^n \frac{1}{n} \rightarrow 0$



求 $\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n^2 + 1}} + \frac{1}{\sqrt{n^2 + 2}} + \cdots + \frac{1}{\sqrt{n^2 + n}} \right).$

数列与函数的关系

THEOREM 3—The Continuous Function Theorem for Sequences Let $\{a_n\}$ be a sequence of real numbers. If $a_n \rightarrow L$ and if f is a function that is continuous at L and defined at all a_n , then $f(a_n) \rightarrow f(L)$.

可以摆脱外函数的限制，直接求带 n 项的极限。

Show that $\sqrt{(n+1)/n} \rightarrow 1$.

THEOREM 4 Suppose that $f(x)$ is a function defined for all $x \geq n_0$ and that $\{a_n\}$ is a sequence of real numbers such that $a_n = f(n)$ for $n \geq n_0$. Then

$$\lim_{x \rightarrow \infty} f(x) = L \quad \Rightarrow \quad \lim_{n \rightarrow \infty} a_n = L.$$

可以使用洛必达法则等求函数极限方法

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0.$$

重要数列极限

THEOREM 5 The following six sequences converge to the limits listed below:

1. $\lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0$

2. $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$

3. $\lim_{n \rightarrow \infty} x^{1/n} = 1 \quad (x > 0)$

4. $\lim_{n \rightarrow \infty} x^n = 0 \quad (|x| < 1)$

5. $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x \quad (\text{any } x)$

6. $\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0 \quad (\text{any } x)$

In Formulas (3) through (6), x remains fixed as $n \rightarrow \infty$.

(b) $\sqrt[n]{n^2}$ (c) $\sqrt[n]{3n}$ (e) $\left(\frac{n-2}{n}\right)^n$ (f) $\frac{100^n}{n!}$

斯特林公式

$$n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$$

or

$$\lim_{n \rightarrow \infty} \frac{n!}{\sqrt{2\pi n} \left(\frac{n}{e}\right)^n} = 1$$

$$\ln \ln n \ll \ln n \ll n^\alpha \ll a^n \ll n! \ll n^n$$

Proof of Stirling's formula requires monotone bounded convergence theorem and

证明?

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

沃利斯公式

Find the limit

$$\lim_{n \rightarrow \infty} \frac{(2n-1)!!}{(2n)!!} = \lim_{n \rightarrow \infty} \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)}$$

Notice that $(2n-1)(2n+1) = 4n^2 - 1 < (2n)^2$,

$$[(2n-1)!!]^2(2n+1) = \prod_{k=1}^n (2k-1)(2k+1) < \prod_{k=1}^n (2k)^2 = ((2n)!!)^2$$

$$0 < \frac{(2n-1)!!}{(2n)!!} < \frac{1}{\sqrt{2n+1}} \rightarrow 0$$

$$\lim_{n \rightarrow \infty} \left(\frac{(2n)!!}{(2n-1)!!} \right)^2 \frac{1}{2n+1} = \frac{\pi}{2}$$

求数列极限（迭代型）

1. 求极限 $\lim_{n \rightarrow \infty} \sqrt[n]{1^n + 2^n + 3^n}$.

2. 设 $0 \leq a_1 \leq a_2 \leq \cdots \leq a_k$

则 $\lim_{n \rightarrow \infty} \sqrt[n]{a_1^n + a_2^n + \cdots + a_k^n} = a_k$

3. $\lim_{n \rightarrow \infty} \sqrt[n]{n^2 - n + 2} =$

4.
$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)}$$

$$\sum_{k=1}^{\infty} \frac{1}{k(k+2)(k+6)}$$

5.

$$\lim_{n \rightarrow \infty} \left(\sqrt{n + \sqrt{n + 2\sqrt{n}}} - \sqrt{n} \right)$$

6.

$$(n!)^{\frac{1}{n^2}}$$

数列的形式

迭代形式 (iterative form) $\longrightarrow$ 递归形式 (recursive form)

$$\frac{\ln(n^2)}{n} = \frac{2 \ln n}{n} \rightarrow 2 \cdot 0 = 0$$

$$\sqrt[n]{n^2} = n^{2/n} = (n^{1/n})^2 \rightarrow (1)^2 = 1$$

$$\sqrt[n]{3n} = 3^{1/n}(n^{1/n}) \rightarrow 1 \cdot 1 = 1$$

$$\left(-\frac{1}{2}\right)^n \rightarrow 0$$

$$\left(\frac{n-2}{n}\right)^n = \left(1 + \frac{-2}{n}\right)^n \rightarrow e^{-2}$$

$$\frac{100^n}{n!} \rightarrow 0$$

$$a_1 = 1 \text{ and } a_n = a_{n-1} + 1$$

$$a_1 = 1, a_2 = 1, \text{ and } a_{n+1} = a_n + a_{n-1}$$

$$a_1 = -4, \quad a_{n+1} = \sqrt{8 + 2a_n}$$

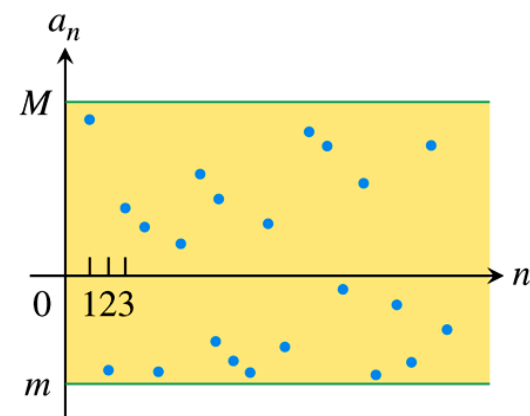
数列的有界性

DEFINITIONS A sequence $\{a_n\}$ is **bounded from above** if there exists a number M such that $a_n \leq M$ for all n . The number M is an **upper bound** for $\{a_n\}$. If M is an upper bound for $\{a_n\}$ but no number less than M is an upper bound for $\{a_n\}$, then M is the **least upper bound** for $\{a_n\}$.

A sequence $\{a_n\}$ is **bounded from below** if there exists a number m such that $a_n \geq m$ for all n . The number m is a **lower bound** for $\{a_n\}$. If m is a lower bound for $\{a_n\}$ but no number greater than m is a lower bound for $\{a_n\}$, then m is the **greatest lower bound** for $\{a_n\}$.

If $\{a_n\}$ is bounded from above and below, the $\{a_n\}$ is **bounded**. If $\{a_n\}$ is not bounded, then we say that $\{a_n\}$ is an **unbounded** sequence.

上界；下界；最小上界；最大下界



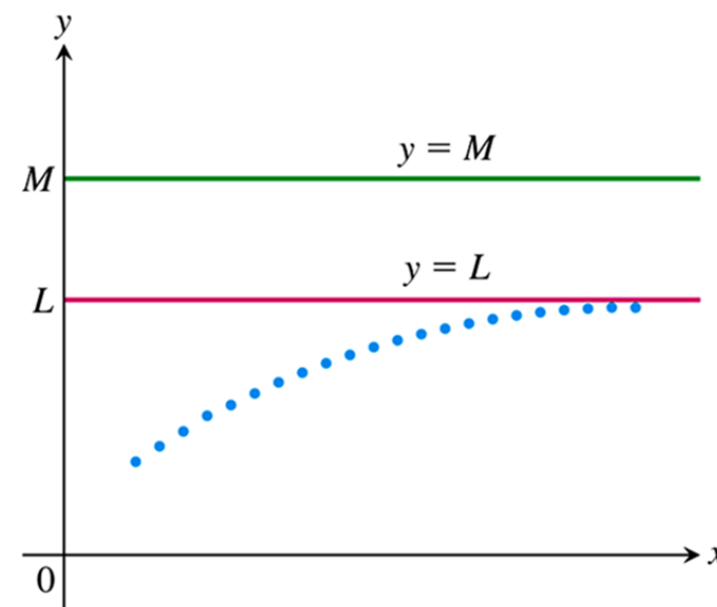
数列的单调性

DEFINITIONS A sequence $\{a_n\}$ is **nondecreasing** if $a_n \leq a_{n+1}$ for all n . That is, $a_1 \leq a_2 \leq a_3 \leq \dots$. The sequence is **nonincreasing** if $a_n \geq a_{n+1}$ for all n . The sequence $\{a_n\}$ is **monotonic** if it is either nondecreasing or nonincreasing.

THEOREM 6—The Monotonic Sequence Theorem If a sequence $\{a_n\}$ is both bounded and monotonic, then the sequence converges.

1. 单调有界推出一个数列收敛
2. 数列收敛不能推出单调或是有界
3. 单调增有上界，单调减有下界

例 证明数列 $x_n = \sqrt{3 + \sqrt{3 + \sqrt{\dots + \sqrt{3}}}}$ (n 重根式) 的极限存在.



求数列极限（递归型）

1.

Assume that sequence $\{a_n\}$ converges, find its limit.

$$a_1 = -4, \quad a_{n+1} = \sqrt{8 + 2a_n}.$$

2.

$x_1 > 0$, $x_{n+1} = 1 + \frac{x_n}{x_n+1}$. Prove that the sequence $\{x_n\}$ is convergent and find its limit.

3.

$a_1 = 3$, $a_{n+1} = 12 - \sqrt{a_n}$. Prove that the sequence $\{a_n\}$ is convergent and find its limit.

无穷级数的定义

DEFINITIONS Given a sequence of numbers $\{a_n\}$, an expression of the form

$$a_1 + a_2 + a_3 + \cdots + a_n + \cdots$$

is an **infinite series**. The number a_n is the **n th term** of the series. The sequence $\{s_n\}$ defined by

$$\begin{aligned} s_1 &= a_1 \\ s_2 &= a_1 + a_2 \\ &\vdots \\ s_n &= a_1 + a_2 + \cdots + a_n = \sum_{k=1}^n a_k \\ &\vdots \end{aligned}$$

is the **sequence of partial sums** of the series, the number s_n being the **n th partial sum**. If the sequence of partial sums converges to a limit L , we say that the series **converges** and that its **sum** is L . In this case, we also write

$$a_1 + a_2 + \cdots + a_n + \cdots = \sum_{n=1}^{\infty} a_n = L.$$

If the sequence of partial sums of the series does not converge, we say that the series **diverges**.

常数项级数收敛(发散)



$\lim_{n \rightarrow \infty} s_n$ 存在(不存在)

好求级数（直接判别法）

Geometric series are series of the form

$$a + ar + ar^2 + \cdots + ar^{n-1} + \cdots = \sum_{n=1}^{\infty} ar^{n-1}$$

in which a and r are fixed real numbers and $a \neq 0$.

几何级数

If $|r| < 1$, the geometric series $a + ar + ar^2 + \cdots + ar^{n-1} + \cdots$ converges to $a/(1 - r)$:

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1 - r}, \quad |r| < 1.$$

If $|r| \geq 1$, the series diverges.

“裂项”级数

the “telescoping” series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$

级数收敛的必要条件（排除法）

THEOREM 7 If $\sum_{n=1}^{\infty} a_n$ converges, then $a_n \rightarrow 0$.

Conversely, if $a_n \rightarrow 0$, $\sum_{n=1}^{\infty} a_n$ may fail to converge.

1. 如果级数的一般项不趋于零, 则级数发散;
2. 必要条件不充分.

The n th-Term Test for Divergence

$\sum_{n=1}^{\infty} a_n$ diverges if $\lim_{n \rightarrow \infty} a_n$ fails to exist or is different from zero.

调和级数 $1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} + \cdots$

$\sum_{n=1}^{\infty} n^2$ diverges because $n^2 \rightarrow \infty$.

$\sum_{n=1}^{\infty} \frac{n+1}{n}$ diverges because $\frac{n+1}{n} \rightarrow 1$. $\lim_{n \rightarrow \infty} a_n \neq 0$

$\sum_{n=1}^{\infty} (-1)^{n+1}$ diverges because $\lim_{n \rightarrow \infty} (-1)^{n+1}$ does not exist.

$\sum_{n=1}^{\infty} \frac{-n}{2n+5}$ diverges because $\lim_{n \rightarrow \infty} \frac{-n}{2n+5} = -\frac{1}{2} \neq 0$.

数列级数的运算法则

THEOREM 8 If $\sum a_n = A$ and $\sum b_n = B$ are convergent series, then

1. *Sum Rule:* $\sum(a_n + b_n) = \sum a_n + \sum b_n = A + B$
2. *Difference Rule:* $\sum(a_n - b_n) = \sum a_n - \sum b_n = A - B$
3. *Constant Multiple Rule:* $\sum ka_n = k\sum a_n = kA$ (any number k).

只会用到加减和数乘！

1. Every nonzero constant multiple of a divergent series diverges.
2. If $\sum a_n$ converges and $\sum b_n$ diverges, then $\sum(a_n + b_n)$ and $\sum(a_n - b_n)$ both diverge.

函数极限的运算法则

THEOREM 1—Limit Laws If L, M, c , and k are real numbers and

$$\lim_{x \rightarrow c} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow c} g(x) = M, \quad \text{then}$$

1. *Sum Rule*: $\lim_{x \rightarrow c} (f(x) + g(x)) = L + M$

2. *Difference Rule*: $\lim_{x \rightarrow c} (f(x) - g(x)) = L - M$

3. *Constant Multiple Rule*: $\lim_{x \rightarrow c} (k \cdot f(x)) = k \cdot L$

4. *Product Rule*: $\lim_{x \rightarrow c} (f(x) \cdot g(x)) = L \cdot M$

5. *Quotient Rule*: $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{L}{M}, \quad M \neq 0$

6. *Power Rule*: $\lim_{x \rightarrow c} [f(x)]^n = L^n, \quad n \text{ a positive integer}$

7. *Root Rule*: $\lim_{x \rightarrow c} \sqrt[n]{f(x)} = \sqrt[n]{L} = L^{1/n}, \quad n \text{ a positive integer}$

(If n is even, we assume that $\lim_{x \rightarrow c} f(x) = L > 0$.)

1. 有 + 无 = 无

2. 无 + 无 = 不定

3. 有 * 无 = 不定

4. 无 * 无 = 不定

加减项与重建索引

We can add a finite number of terms to a series or delete a finite number of terms without altering the series' convergence or divergence, although in the case of convergence this will usually change the sum.

$$\sum_{n=1}^{\infty} \frac{1}{5^n} = \frac{1}{5} + \frac{1}{25} + \frac{1}{125} + \sum_{n=4}^{\infty} \frac{1}{5^n}$$

and

$$\sum_{n=4}^{\infty} \frac{1}{5^n} = \left(\sum_{n=1}^{\infty} \frac{1}{5^n} \right) - \frac{1}{5} - \frac{1}{25} - \frac{1}{125}.$$

As long as we preserve the order of its terms, we can reindex any series without altering its convergence.

$$a_1 + a_2 + a_3 + \cdots = \sum_{n=1}^{\infty} a_n = \sum_{n=1+h}^{\infty} a_{n-h} = \sum_{n=1-h}^{\infty} a_{n+h}$$

加减有限的项
不会改变级数敛散性

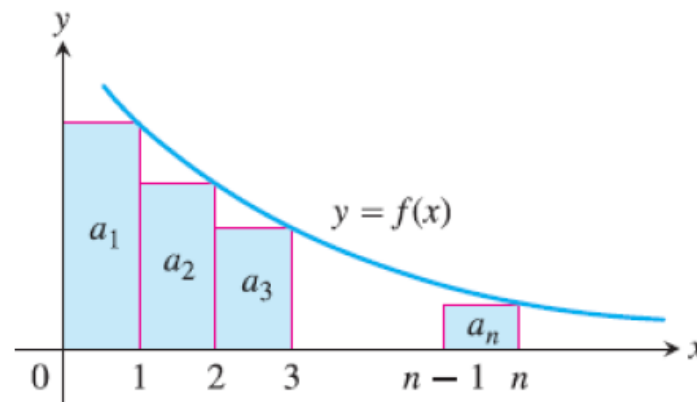
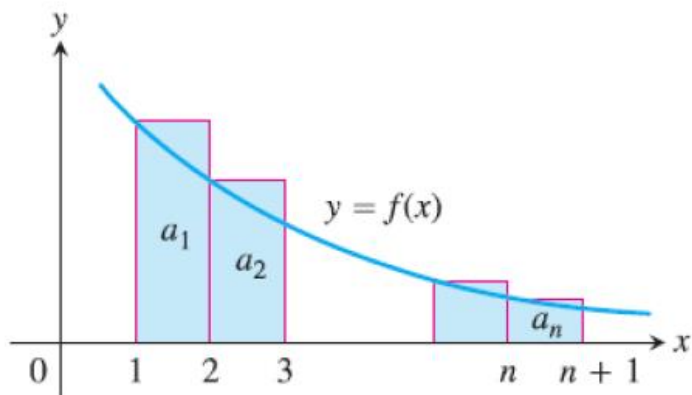
项的次序不变
不会改变级数敛散性

积分判别法

THEOREM 9—The Integral Test Let $\{a_n\}$ be a sequence of positive terms. Suppose that $a_n = f(n)$, where f is a continuous, positive, decreasing function of x for all $x \geq N$ (N a positive integer). Then the series $\sum_{n=N}^{\infty} a_n$ and the integral $\int_N^{\infty} f(x) dx$ both converge or both diverge.

数列级数敛散性 $\iff$ 函数广义积分敛散性

证明?



$$\int_1^{n+1} f(x) dx \leq a_1 + a_2 + \cdots + a_n \leq a_1 + \int_1^n f(x) dx.$$

P级数

EXAMPLE 3 Show that the p -series

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \cdots + \frac{1}{n^p} + \cdots$$

(p a real constant) converges if $p > 1$, and diverges if $p \leq 1$.

$$\frac{1}{n^{\alpha} (\ln n)^{\beta}} \begin{cases} \text{converge} & \text{if } \alpha > 1 \text{ or } \alpha = 1 \ \beta > 1 \\ \text{diverge} & \text{if } \alpha < 1 \text{ or } \alpha = 1 \ \beta \leq 1 \end{cases}$$

证明?

(a) $\sum_{n=1}^{\infty} n e^{-n^2}$

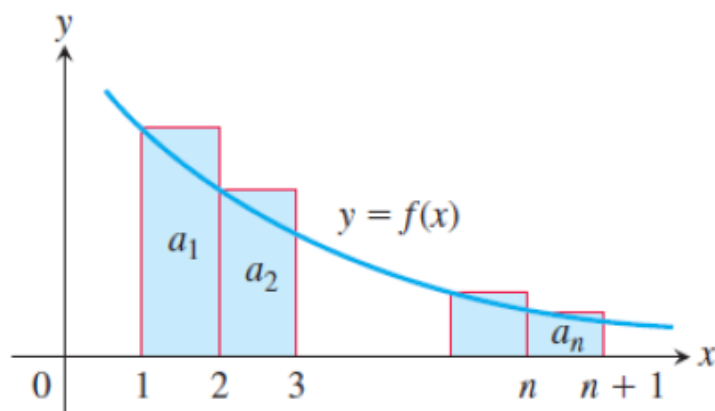
(b) $\sum_{n=1}^{\infty} \frac{1}{2^{\ln n}}$

余数的界限

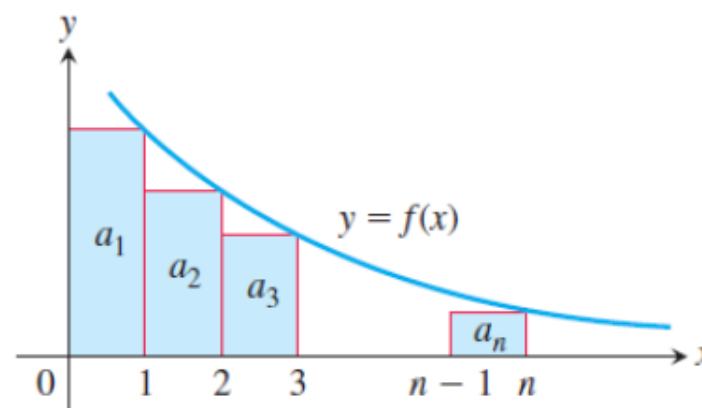
Bounds for the Remainder in the Integral Test

Suppose $\{a_k\}$ is a sequence of positive terms with $a_k = f(k)$, where f is a continuous positive decreasing function of x for all $x \geq n$, and that $\sum a_n$ converges to S . Then the remainder $R_n = S - s_n$ satisfies the inequalities

$$\int_{n+1}^{\infty} f(x) dx \leq R_n \leq \int_n^{\infty} f(x) dx. \quad (1)$$



(a)



(b)

直接比较判别法

THEOREM 10—The Comparison Test Let $\sum a_n$, $\sum c_n$, and $\sum d_n$ be series with nonnegative terms. Suppose that for some integer N

$$d_n \leq a_n \leq c_n \quad \text{for all } n > N.$$

- (a) If $\sum c_n$ converges, then $\sum a_n$ also converges.
- (b) If $\sum d_n$ diverges, then $\sum a_n$ also diverges.

the partial sums of $\sum a_n$ are bounded above by

$$M = a_1 + a_2 + \cdots + a_N + \sum_{n=N+1}^{\infty} c_n.$$

the partial sums for $\sum d_n$ would be bounded by

$$M^* = d_1 + d_2 + \cdots + d_N + \sum_{n=N+1}^{\infty} a_n$$

$$\sum_{n=1}^{\infty} \frac{5}{5n-1}$$

$$\frac{1}{2^n + \sqrt{n}}$$

证明?

极限比较判别法

THEOREM 11—Limit Comparison Test

all $n \geq N$ (N an integer).

Suppose that $a_n > 0$ and $b_n > 0$ for

1. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c > 0$, then $\sum a_n$ and $\sum b_n$ both converge or both diverge.
2. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges.
3. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

Does $\sum_{n=1}^{\infty} \frac{\ln n}{n^{3/2}}$ converge?

几何级数/等比级数

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}, \quad |r| < 1.$$

p级数（或 ln 形式）

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \cdots + \frac{1}{n^p} + \cdots$$

敛散性判别 *

THEOREM 2—Direct Comparison Test Let f and g be continuous on $[a, \infty)$ with $0 \leq f(x) \leq g(x)$ for all $x \geq a$. Then

1. $\int_a^\infty f(x) dx$ converges if $\int_a^\infty g(x) dx$ converges.

2. $\int_a^\infty g(x) dx$ diverges if $\int_a^\infty f(x) dx$ diverges.

判断: 1. $\int_1^\infty \frac{\arctan x}{(1+x^2)^{\frac{3}{2}}} dx$

2. $\int_1^\infty x^{p-1} e^{-x} dx$ 敛散性

3. $\int_0^1 \frac{1}{\sqrt[4]{1-x^4}} dx$

THEOREM 3—Limit Comparison Test If the positive functions f and g are continuous on $[a, \infty)$, and if

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L, \quad 0 < L < \infty,$$

then

$$\int_a^\infty f(x) dx \quad \text{and} \quad \int_a^\infty g(x) dx$$

both converge or both diverge.

用p级数比较非常重要!

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L \quad (\text{positive})$$

$$\begin{cases} L = \infty & f(x) \text{ 收敛则 } g(x) \text{ 同} \\ 0 < L < \infty & \text{同敛散} \\ L = 0 & g(x) \text{ 收敛则 } f(x) \text{ 同} \end{cases}$$

判别级数敛散性（比较审敛）

判别 $\sum_{n=1}^{\infty} (1 - \cos \frac{x}{n})$ 的敛散性

(1) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^2 + n}}$

(3) $\sum_{n=1}^{\infty} \frac{1}{1 + a^n} \quad (a > 0)$

(2) $\sum_{n=1}^{\infty} \sin \frac{1}{n},$

$$\sum_{n=1}^{\infty} \frac{n+1}{n^2 - n + 3},$$

级数绝对收敛 (Absolutely)

只含非负项的级数 $\implies$ 具有正负项的级数

DEFINITION A series $\sum a_n$ **converges absolutely** (is **absolutely convergent**) if the corresponding series of absolute values, $\sum |a_n|$, converges.

THEOREM 12—The Absolute Convergence Test If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.

级数绝对收敛 $\implies$ 级数收敛

For each n ,

证明?

$$-|a_n| \leq a_n \leq |a_n|, \quad \text{so} \quad 0 \leq a_n + |a_n| \leq 2|a_n|.$$

Ratio判别法与Root判别法

THEOREM 13—The Ratio Test Let $\sum a_n$ be any series and suppose that

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \rho.$$

Then (a) the series *converges absolutely* if $\rho < 1$, (b) the series *diverges* if $\rho > 1$ or ρ is infinite, (c) the test is *inconclusive* if $\rho = 1$.

THEOREM 14—The Root Test Let $\sum a_n$ be any series and suppose that

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \rho.$$

Then (a) the series *converges absolutely* if $\rho < 1$, (b) the series *diverges* if $\rho > 1$ or ρ is infinite, (c) the test is *inconclusive* if $\rho = 1$.

1. 更适用于带有阶乘, n 次方或者连乘号的级数
2. 不必找参考级数

Consider again the series with terms $a_n = \begin{cases} n/2^n, & n \text{ odd} \\ 1/2^n, & n \text{ even.} \end{cases}$

交替级数

A series in which the terms are alternately positive and negative is an **alternating series**.

examples:

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \cdots + \frac{(-1)^{n+1}}{n} + \cdots \quad (1)$$

$$-2 + 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \cdots + \frac{(-1)^n 4}{2^n} + \cdots \quad (2)$$

the n th term of an alternating series is of the form

$$a_n = (-1)^{n+1} u_n \quad \text{or} \quad a_n = (-1)^n u_n$$

where $u_n = |a_n|$ is a positive number.

正负交替的级数为交替级数

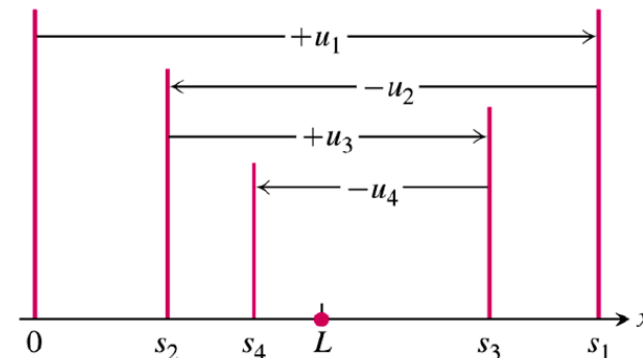
莱布尼茨判别法

THEOREM 14—The Alternating Series Test (Leibniz's Test) The series

$$\sum_{n=1}^{\infty} (-1)^{n+1} u_n = u_1 - u_2 + u_3 - u_4 + \cdots$$

converges if all three of the following conditions are satisfied:

1. The u_n 's are all positive.
2. The positive u_n 's are (eventually) nonincreasing: $u_n \geq u_{n+1}$ for all $n \geq N$, for some integer N .
3. $u_n \rightarrow 0$.



THEOREM 15—The Alternating Series Estimation Theorem If the alternating series $\sum_{n=1}^{\infty} (-1)^{n+1} u_n$ satisfies the three conditions of Theorem 14, then for $n \geq N$,

$$s_n = u_1 - u_2 + \cdots + (-1)^{n+1} u_n$$

approximates the sum L of the series with an error whose absolute value is less than u_{n+1} , the absolute value of the first unused term. Furthermore, the sum L lies between any two successive partial sums s_n and s_{n+1} , and the remainder, $L - s_n$, has the same sign as the first unused term.

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n}$$

$$\sum_{n=0}^{\infty} (-1)^n \frac{1}{2^n}$$

级数条件收敛 (Conditionally)

DEFINITION A series that converges but does not converge absolutely **converges conditionally**.

级数绝对收敛 VS. 级数条件收敛?

本质区别: 是否满足重排定理。

THEOREM 17—The Rearrangement Theorem for Absolutely Convergent Series If $\sum_{n=1}^{\infty} a_n$ converges absolutely, and $b_1, b_2, \dots, b_n, \dots$ is any arrangement of the sequence $\{a_n\}$, then $\sum b_n$ converges absolutely and

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} a_n.$$

设正项级数 $\sum_{n=1}^{\infty} u_n$ 收敛, 能否推得 $\sum_{n=1}^{\infty} u_n^2$ 收敛? 反之是否成立?

设级数 $\sum_{n=1}^{\infty} u_n$ 收敛, 能否推得 $\sum_{n=1}^{\infty} u_n^2$ 收敛? 反之是否成立?

敛散性判别

If $0 < a_n < \frac{1}{n}$, which series must be convergent?

$$\sum_{n=1}^{\infty} a_n, \quad \sum_{n=1}^{\infty} (-1)^n a_n, \quad \sum_{n=1}^{\infty} \sqrt{a_n}, \quad \sum_{n=1}^{\infty} \frac{a_n}{n}$$

If the series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent, prove that $\sum_{n=1}^{\infty} \frac{a_n}{1 + a_n}$ is also absolutely convergent.

$a \in \mathbb{R}$, $\sum_{n=1}^{\infty} \left(\frac{\sin a}{n^2} - \frac{1}{\sqrt{n}} \right)$ is divergent.

Justify the convergence of

$$\sum_{n=2}^{\infty} \frac{(-1)^n}{\sqrt{n + (-1)^n}}$$

幂级数

DEFINITIONS A power series about $x = 0$ is a series of the form

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n + \cdots \quad (1)$$

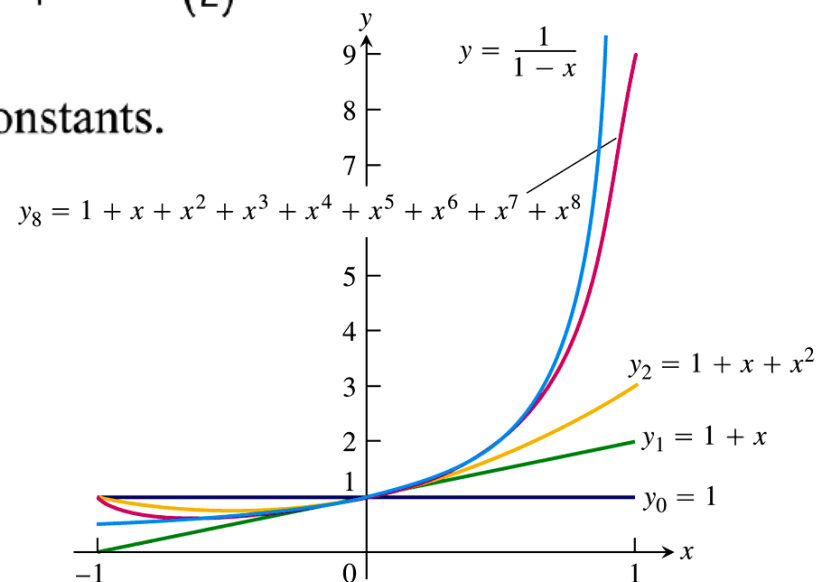
A power series about $x = a$ is a series of the form

$$\sum_{n=0}^{\infty} c_n (x - a)^n = c_0 + c_1 (x - a) + c_2 (x - a)^2 + \cdots + c_n (x - a)^n + \cdots \quad (2)$$

in which the **center** a and the **coefficients** $c_0, c_1, c_2, \dots, c_n, \dots$ are constants.

几何级数:

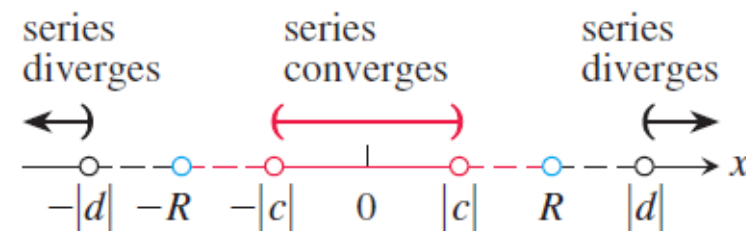
$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \cdots + x^n + \cdots$$



幂级数收敛定理

THEOREM 18—The Convergence Theorem for Power Series If the power series

$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots$ converges at $x = c \neq 0$, then it converges absolutely for all x with $|x| < |c|$. If the series diverges at $x = d$, then it diverges for all x with $|x| > |d|$.



COROLLARY TO THEOREM 18 The convergence of the series $\sum c_n(x - a)^n$ is described by one of the following three cases:

1. There is a positive number R such that the series diverges for x with $|x - a| > R$ but converges absolutely for x with $|x - a| < R$. The series may or may not converge at either of the endpoints $x = a - R$ and $x = a + R$.
2. The series converges absolutely for every x ($R = \infty$).
3. The series converges at $x = a$ and diverges elsewhere ($R = 0$).

1. 用判断方法划定收敛范围（找到R）
2. $|R|$ 内部绝对收敛
3. $|R|$ 外部发散
4. 对于R点（端点值）单独判断

幂级数收敛定理

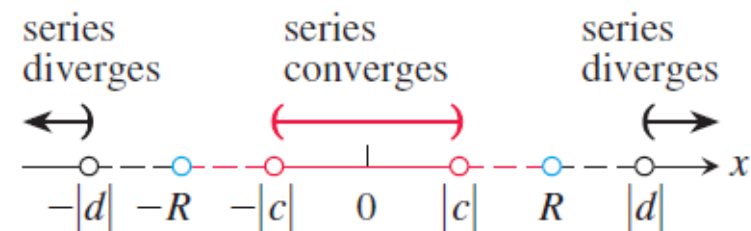
How to Test a Power Series for Convergence

1. Use the **Ratio Test (or Root Test)** to find the interval where the series converges absolutely. Ordinarily, this is an open interval

$$|x - a| < R \quad \text{or} \quad a - R < x < a + R.$$

2. If the interval of absolute convergence is finite, test for convergence or divergence at each endpoint, as in Examples 3a and b. Use a **Comparison Test, the Integral Test, or the Alternating Series Test.**
3. If the interval of absolute convergence is $a - R < x < a + R$, the series diverges for $|x - a| > R$ (it does not even converge conditionally) because the n th term does not approach zero for those values of x .

求幂级数 $\sum_{n=1}^{\infty} \frac{x^{2n-1}}{2^n}$ 的收敛区间.



1. 用判断方法划定收敛范围（找到R）
2. $|R|$ 内部绝对收敛
3. $|R|$ 外部发散
4. 对于R点（端点值）单独判断

幂级数的运算法则

设 $\sum_{n=0}^{\infty} a_n x^n$ 和 $\sum_{n=0}^{\infty} b_n x^n$ 的收敛半径各为 R_1 和 R_2 , $R = \min \{R_1, R_2\}$

(1) 加减法

$$\sum_{n=0}^{\infty} a_n x^n \pm \sum_{n=0}^{\infty} b_n x^n = \sum_{n=0}^{\infty} c_n x^n \quad x \in (-R, R) \quad c_n = a_n \pm b_n$$

(2) 乘法

THEOREM 19—The Series Multiplication Theorem for Power Series If $A(x) = \sum_{n=0}^{\infty} a_n x^n$ and $B(x) = \sum_{n=0}^{\infty} b_n x^n$ converge absolutely for $|x| < R$, and

$$c_n = a_0 b_n + a_1 b_{n-1} + a_2 b_{n-2} + \cdots + a_{n-1} b_1 + a_n b_0 = \sum_{k=0}^n a_k b_{n-k},$$

then $\sum_{n=0}^{\infty} c_n x^n$ converges absolutely to $A(x)B(x)$ for $|x| < R$:

$$\left(\sum_{n=0}^{\infty} a_n x^n \right) \cdot \left(\sum_{n=0}^{\infty} b_n x^n \right) = \sum_{n=0}^{\infty} c_n x^n.$$

幂级数的衍生函数敛散性

THEOREM 20 If $\sum_{n=0}^{\infty} a_n x^n$ converges absolutely for $|x| < R$, then $\sum_{n=0}^{\infty} a_n (f(x))^n$ converges absolutely for any continuous function f on $|f(x)| < R$.

THEOREM 21—The Term-by-Term Differentiation Theorem If $\sum c_n(x - a)^n$ has radius of convergence $R > 0$, it defines a function

$$f(x) = \sum_{n=0}^{\infty} c_n(x - a)^n \quad \text{on the interval} \quad a - R < x < a + R.$$

This function f has derivatives of all orders inside the interval, and we obtain the derivatives by differentiating the original series term by term:

$$f'(x) = \sum_{n=1}^{\infty} n c_n(x - a)^{n-1},$$

$$f''(x) = \sum_{n=2}^{\infty} n(n-1) c_n(x - a)^{n-2},$$

and so on. Each of these derived series converges at every point of the interval $a - R < x < a + R$.

THEOREM 22—The Term-by-Term Integration Theorem Suppose that

$$f(x) = \sum_{n=0}^{\infty} c_n(x - a)^n$$

converges for $a - R < x < a + R$ ($R > 0$). Then

$$\sum_{n=0}^{\infty} c_n \frac{(x - a)^{n+1}}{n + 1}$$

converges for $a - R < x < a + R$ and

$$\int f(x) dx = \sum_{n=0}^{\infty} c_n \frac{(x - a)^{n+1}}{n + 1} + C$$

for $a - R < x < a + R$.

幂级数逐项求导后，收敛半径不变，但它的收敛域的端点可能改变（越求导越小）。

求幂级数

例 求 $\sum_{n=1}^{\infty} \frac{n(n+1)}{2^n}$ 的和.

求幂级数 $\sum_{n=0}^{\infty} \left(\frac{4n^2 + 4n + 3}{2n + 1} \right) x^{2n}$ 的和函数.

Find the radius and interval of convergence of the power series

$$\sum_{n=1}^{\infty} \left(1 + \frac{1}{2} + \cdots + \frac{1}{n} \right) x^n$$

泰勒级数及麦克劳林级数

DEFINITIONS Let f be a function with derivatives of all orders throughout some interval containing a as an interior point. Then the **Taylor series generated by f at $x = a$** is

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x - a)^k = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!} (x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!} (x - a)^n + \cdots.$$

The **Maclaurin series generated by f** is

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k = f(0) + f'(0)x + \frac{f''(0)}{2!} x^2 + \cdots + \frac{f^{(n)}(0)}{n!} x^n + \cdots,$$

the Taylor series generated by f at $x = 0$.

敛散性？

麦克劳林级数就是在 $x = 0$ 处的泰勒级数

泰勒多项式与泰勒定理

DEFINITION Let f be a function with derivatives of order k for $k = 1, 2, \dots, N$ in some interval containing a as an interior point. Then for any integer n from 0 through N , the **Taylor polynomial of order n** generated by f at $x = a$ is the polynomial

$$P_n(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \dots \\ + \frac{f^{(k)}(a)}{k!}(x - a)^k + \dots + \frac{f^{(n)}(a)}{n!}(x - a)^n.$$

缺少后面项的泰勒级数

THEOREM 23—Taylor's Theorem If f and its first n derivatives $f', f'', \dots, f^{(n)}$ are continuous on the closed interval between a and b , and $f^{(n)}$ is differentiable on the open interval between a and b , then there exists a number c between a and b such that

$$f(b) = f(a) + f'(a)(b - a) + \frac{f''(a)}{2!}(b - a)^2 + \dots \\ + \frac{f^{(n)}(a)}{n!}(b - a)^n + \boxed{\frac{f^{(n+1)}(c)}{(n+1)!}(b - a)^{n+1}}.$$

将后面项用一个残差项替代

泰勒公式与残差项

Taylor's Formula

If f has derivatives of all orders in an open interval I containing a , then for each positive integer n and for each x in I ,

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x - a)^n + R_n(x), \quad (1)$$

泰勒公式 = 泰勒多项式 + 残差项

where

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x - a)^{n+1} \quad \text{for some } c \text{ between } a \text{ and } x. \quad (2)$$

If $R_n(x) \rightarrow 0$ as $n \rightarrow \infty$ for all $x \in I$, we say that the Taylor series generated by f at $x = a$ **converges** to f on I , and we write

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x - a)^k.$$

残差项趋于零，泰勒级数收敛

估算残差项

THEOREM 24—The Remainder Estimation Theorem If there is a positive constant M such that $|f^{(n+1)}(t)| \leq M$ for all t between x and a , inclusive, then the remainder term $R_n(x)$ in Taylor's Theorem satisfies the inequality

$$|R_n(x)| \leq M \frac{|x - a|^{n+1}}{(n + 1)!}.$$

If this inequality holds for every n and the other conditions of Taylor's Theorem are satisfied by f , then the series converges to $f(x)$.

Show that the Taylor series for $\sin x$ at $x = 0$ converges for all x .

二次项级数

The Binomial Series

For $-1 < x < 1$,

$$(1 + x)^m = 1 + \sum_{k=1}^{\infty} \binom{m}{k} x^k,$$

where we define

$$\binom{m}{1} = m, \quad \binom{m}{2} = \frac{m(m-1)}{2!},$$

and

$$\binom{m}{k} = \frac{m(m-1)(m-2)\cdots(m-k+1)}{k!} \quad \text{for } k \geq 3.$$

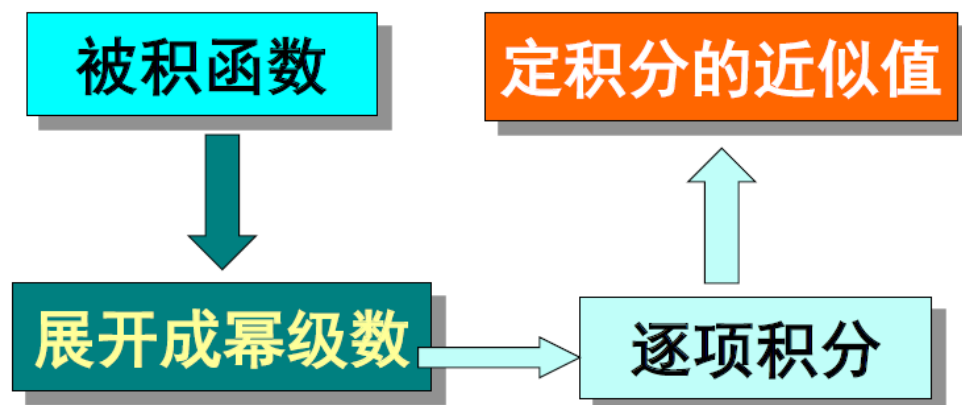
记住这个形式，很多时候可以直接展开来解决很多问题！

泰勒级数应用——近似计算

两类问题：

1. 给定项数，求近似值并估计精度； 2. 给出精度，确定项数.

关键：通过估计余项，确定精度或项数.



$$e^{-x^2}, \frac{\sin x}{x}, \frac{1}{\ln x},$$

泰勒级数应用——求极限

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + \cdots = \sum_{n=0}^{\infty} x^n, \quad |x| < 1$$

$$\frac{1}{1+x} = 1 - x + x^2 - \cdots + (-x)^n + \cdots = \sum_{n=0}^{\infty} (-1)^n x^n, \quad |x| < 1$$

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad |x| < \infty$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad |x| < \infty$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + (-1)^n \frac{x^{2n}}{(2n)!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \quad |x| < \infty$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + (-1)^{n-1} \frac{x^n}{n} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n}, \quad -1 < x \leq 1$$

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots + (-1)^n \frac{x^{2n+1}}{2n+1} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}, \quad |x| \leq 1$$

$$\lim_{x \rightarrow 0} \frac{\sin x - \tan x}{x^3}.$$

$$\text{Find } \lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right).$$

$$\lim_{x \rightarrow 1} \frac{\ln x}{x-1}.$$

Taylor展开、洛必达法则为求极限最powerful的工具

欧拉公式——数学最美的公式

DEFINITION

For any real number θ , $e^{i\theta} = \cos \theta + i \sin \theta$. (4)

$$e^{i\pi} = -1.$$

集两个自然常数，无理数与有理数为一体的公式

间接法求泰勒级数

变量代换, 四则运算, 恒等变形, 逐项求导, 逐项积分

$$\cos x = (\sin x)'$$

将 $f(x) = \sin x \cos 2x$ 展开成 x 的幂级数.

$$\arctan x = \int_0^x \frac{dx}{1+x^2}$$

将 $f(x) = \frac{x-1}{4-x}$ 在 $x=1$ 处展开成泰勒级数

$$\ln(1+x) = \int_0^x \frac{dx}{1+x}$$

Find the Taylor series for $f(x) = \ln(x + \sqrt{x^2 + 1})$ at $x = 0$.

请写出函数 $f(x) = \ln(x + \sqrt{x^2 + 1})$ 在 $x = 0$ 处的 Taylor 级数.

(2019 Midterm exam)

Chapter11

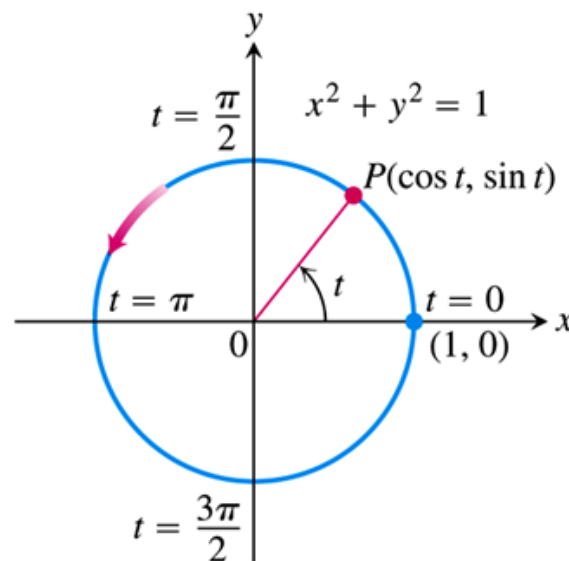
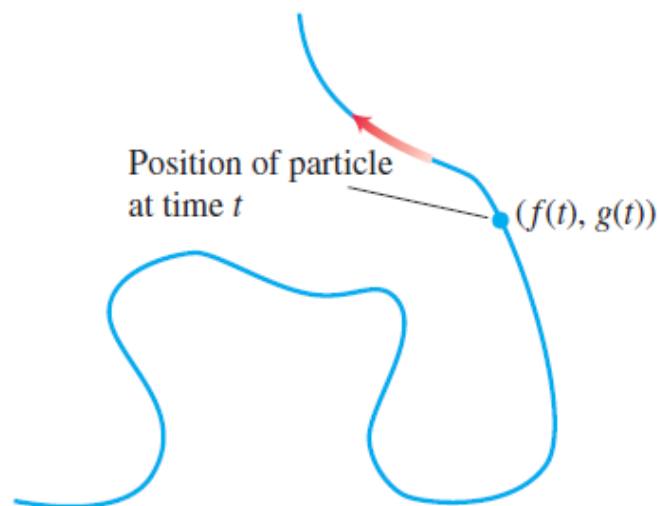
参数方程与极坐标

参数方程

DEFINITION If x and y are given as functions

$$x = f(t), \quad y = g(t)$$

over an interval I of t -values, then the set of points $(x, y) = (f(t), g(t))$ defined by these equations is a **parametric curve**. The equations are **parametric equations** for the curve.



$$x = a \cos t,$$

$$y = a \sin t,$$

$$0 \leq t \leq 2\pi.$$

参数方程微分

Parametric Formula for dy/dx

If all three derivatives exist and $dx/dt \neq 0$,

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} . \quad (1)$$

Parametric Formula for d^2y/dx^2

If the equations $x = f(t)$, $y = g(t)$ define y as a twice-differentiable function of x , then at any point where $dx/dt \neq 0$ and $y' = dy/dx$,

$$\frac{d^2y}{dx^2} = \frac{dy'/dt}{dx/dt} . \quad (2)$$

注意两公式的“联动”

Find d^2y/dx^2 as a function of t if $x = t - t^2$, $y = t - t^3$.

弧长公式 **Arc Length**

DEFINITION If a curve C is defined parametrically by $x = f(t)$ and $y = g(t)$, $a \leq t \leq b$, where f' and g' are continuous and not simultaneously zero on $[a, b]$, and C is traversed exactly once as t increases from $t = a$ to $t = b$, then **the length of C** is the definite integral

$$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2} dt.$$

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

Length of a Curve $y = f(x)$

$$x = t \quad \text{and} \quad y = f(t), \quad a \leq t \leq b,$$

The Arc Length Differential

$$ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

求弧长 *

DEFINITION If f' is continuous on $[a, b]$, then the **length (arc length)** of the curve $y = f(x)$ from the point $A = (a, f(a))$ to the point $B = (b, f(b))$ is the value of the integral

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx. \quad (3)$$

Formula for the Length of $x = g(y)$, $c \leq y \leq d$

If g' is continuous on $[c, d]$, the length of the curve $x = g(y)$ from $A = (g(c), c)$ to $B = (g(d), d)$ is

$$L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy = \int_c^d \sqrt{1 + [g'(y)]^2} dy. \quad (4)$$

旋转体表面积公式

Area of Surface of Revolution for Parametrized Curves

If a smooth curve $x = f(t), y = g(t), a \leq t \leq b$, is traversed exactly once as t increases from a to b , then the areas of the surfaces generated by revolving the curve about the coordinate axes are as follows.

1. Revolution about the x -axis ($y \geq 0$):

$$S = \int_a^b 2\pi y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \quad (5)$$

2. Revolution about the y -axis ($x \geq 0$):

$$S = \int_a^b 2\pi x \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \quad (6)$$

求旋转体的表面积 *

DEFINITION If the function $f(x) \geq 0$ is continuously differentiable on $[a, b]$, the **area of the surface** generated by revolving the graph of $y = f(x)$ about the x -axis is

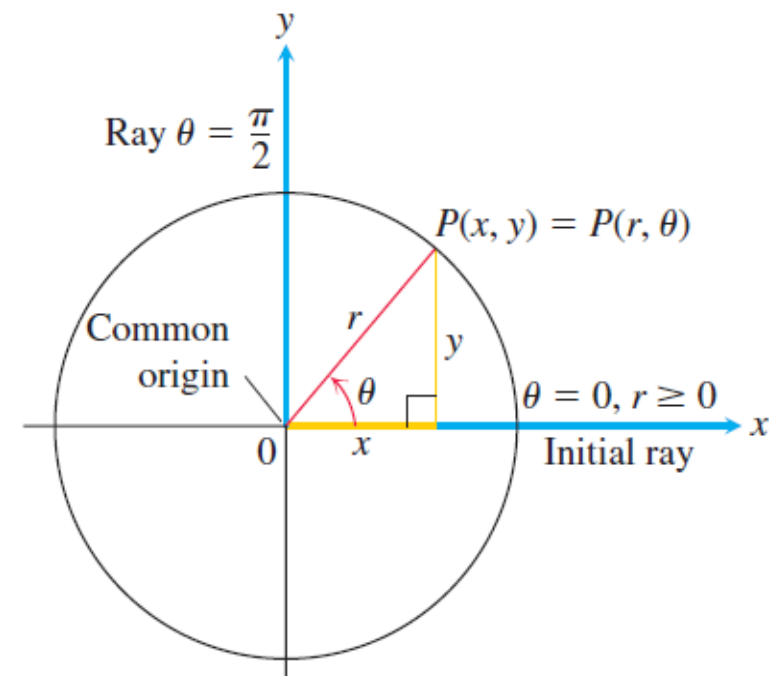
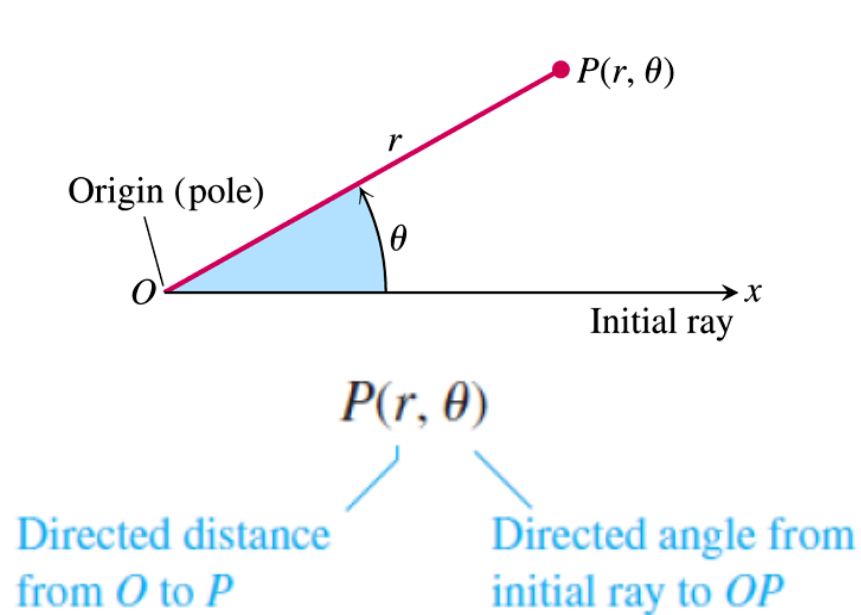
$$S = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_a^b 2\pi f(x) \sqrt{1 + (f'(x))^2} dx. \quad (3)$$

Surface Area for Revolution About the y -Axis

If $x = g(y) \geq 0$ is continuously differentiable on $[c, d]$, the area of the surface generated by revolving the graph of $x = g(y)$ about the y -axis is

$$S = \int_c^d 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy = \int_c^d 2\pi g(y) \sqrt{1 + (g'(y))^2} dy. \quad (4)$$

极坐标



Equations Relating Polar and Cartesian Coordinates

$$x = r \cos \theta, \quad y = r \sin \theta, \quad r^2 = x^2 + y^2, \quad \tan \theta = \frac{y}{x}$$

极坐标微分

Slope of the Curve $r = f(\theta)$

$$\left. \frac{dy}{dx} \right|_{(r, \theta)} = \frac{f'(\theta) \sin \theta + f(\theta) \cos \theta}{f'(\theta) \cos \theta - f(\theta) \sin \theta},$$

provided $dx/d\theta \neq 0$ at (r, θ) .

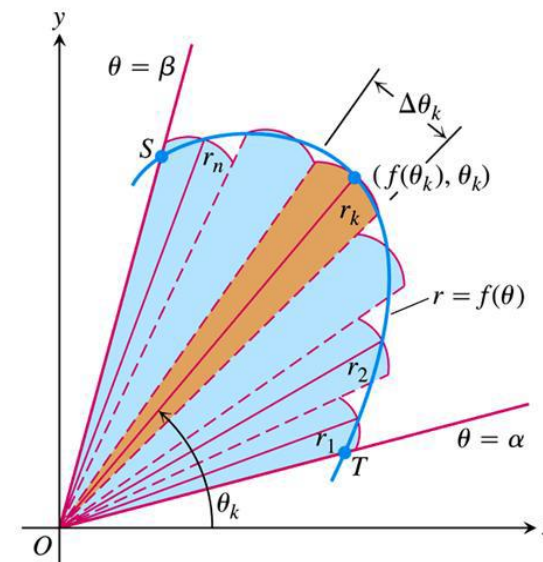
扇形区域面积公式

Area of the Fan-Shaped Region Between the Origin and the Curve $r = f(\theta)$,
 $\alpha \leq \theta \leq \beta$

$$A = \int_{\alpha}^{\beta} \frac{1}{2} r^2 d\theta$$

This is the integral of the **area differential** (Figure 11.32)

$$dA = \frac{1}{2} r^2 d\theta = \frac{1}{2} (f(\theta))^2 d\theta.$$



Area of the Region $0 \leq r_1(\theta) \leq r \leq r_2(\theta)$, $\alpha \leq \theta \leq \beta$

$$A = \int_{\alpha}^{\beta} \frac{1}{2} r_2^2 d\theta - \int_{\alpha}^{\beta} \frac{1}{2} r_1^2 d\theta = \int_{\alpha}^{\beta} \frac{1}{2} (r_2^2 - r_1^2) d\theta \quad (1)$$

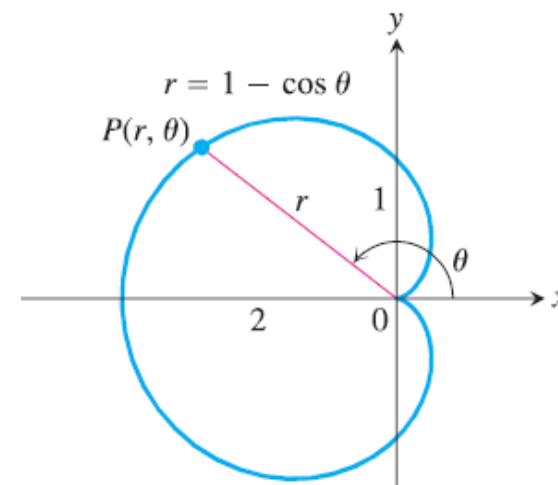
弧长公式

Length of a Polar Curve

If $r = f(\theta)$ has a continuous first derivative for $\alpha \leq \theta \leq \beta$ and if the point $P(r, \theta)$ traces the curve $r = f(\theta)$ exactly once as θ runs from α to β , then the length of the curve is

$$L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta. \quad (3)$$

Find the length of the cardioid $r = 1 - \cos \theta$.





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The background of the slide features a complex network of interconnected nodes and lines, forming a mesh-like structure. The nodes are represented by small orange dots, and the lines are thin orange lines. The background is a light gray color with a subtle pattern of larger, interconnected triangles and polygons.

Q&A

祝大家期中顺利！